

Fish: are vegetarians and vegans missing out by not eating it?

[John Livesey](#) PhD
Scientific Officer
[Department of Endocrinology](#)
[Christchurch Hospital](#)
New Zealand

Dietary 'authorities' often recommend fish as an important part of a healthy diet, usually on the grounds that it improves heart health. For example, the first of the New Zealand National Heart Foundation's *Nine Steps to Eating for a Healthy Heart* reads "Enjoy three meals a day, selecting from dishes that encourage you to eat plant foods and fish, with little or no dairy fat, meat fat or deep fried foods". In a similar vein, the Harvard University Healthy Eating Pyramid includes fish as a positive feature of a healthy diet. The theory is that the large amounts of the long-chain omega-3 fatty acids EPA and/or DHA that are found in fish oil are protective against heart disease.

These sorts of assertions have caused considerable angst among some vegetarians concerned about the health of their heart, because EPA and DHA do not occur in plants, and dairy and eggs contain only very small amounts compared to fish. Now in case you are wondering, we humans do in fact make EPA and DHA ourselves. We make it by carbon chain extension from alpha-linolenic acid, an essential short-chain fatty acid that occurs in plants, most notably linseed (flaxseed), walnuts, canola oil and soya oil. So the assertion that fish is good for you amounts to the claim that we do not make enough EPA and DHA ourselves and would benefit from a dietary source as well.

This raises two questions: would it be possible for everyone to get the suggested supplemental amounts of EPA and DHA from fish? And, if so, would it be desirable? A recent article¹ in the Canadian Medical Association Journal has addressed these questions. The authors include prominent Canadian medical researchers, fisheries scientists and the well-known conservationist Farley Mowat.

They point out that since the global fish catch has been declining since the late 1980's it is physically impossible for everyone in the World to increase their fish consumption. Worse, in the attempt to maintain current fish catches, fish stocks are collapsing at an increasing rate, so much so that the authors predict that all commercially exploited fish stocks will have collapsed by 2050. Hence the typical recommendation to eat two meals of oily fish per week is simply hastening an ecological catastrophe and could not possibly be universally adopted.

Well, if eating fish is not going to be possible (except maybe for the unethical wealthy), how strong is the evidence that lack of fish in the diet harms human health? The Canadian authors reviewed the evidence in their paper; and when all clinical trials, not just the favourable ones, are considered, they show that the supposed benefits are far from clear cut. Indeed, in one trial, the patients given fish oil capsules died at a greater rate!

Tellingly, they also note that vegetarians, who of course consume no fish at all, do not have a greater risk of heart disease, or death in general, than those who eat fish. Which strongly suggests that fish consumption is not essential for optimal health. Indeed, it is generally observed that vegetarians have a lower rate of heart disease than meat-eaters, and the studies of ex-pat Kiwi, Professor Gary Fraser are a good example of this. He also found in his Californian Adventist study subjects that eating fish confers no significant protection from heart disease. Another of his findings was that vegetarians live about two years longer than non-vegetarians, again strongly suggesting that neither fish nor long-chain omega-3's are essential in the diet.

In his book *The Blue Zones*², Dan Buettner identifies four regions of the world where the inhabitants are exceptionally long lived. These are in the mountains of Sardinia, in Okinawa, in Loma Linda in California (the Adventists), and in the Nicoya Peninsula in Costa Rica. In none of these regions is fish a significant part of the diet and the Californian Adventists, being vegetarian, do not eat it at all. So, on this evidence, it seems unlikely that fish consumption is required for longevity. Indeed, there is better evidence that legumes contribute to longevity³.

Fish: are vegetarians and vegans missing out by not eating it?

These observations are both supported by an analysis of the Greek arm of the EPIC study. Legume consumption was found to contribute positively to longevity but fish consumption was found to be without a significant effect. Indeed, fish showed a nonsignificant negative effect on longevity⁴. Further support is given by the finding in an American study of an increased risk of type 2 diabetes in those eating the most fish⁵.

Evolutionary considerations also make it unlikely that fish are required for optimal health. Humans are generally believed to have been evolving on the East African savannah for a million years or so before spreading out to the rest of the world less than 100,000 years ago. But the savannah does not contain oily fish since these are only found in the deep sea, so it seems unlikely that we would have evolved to have a requirement for a food that would not have been available.

It is sometimes suggested that growth and development in children might be adversely affected by absence of long-chain omega-3 fatty acids from the diet. However a British study found that in both men and women, there were no significant differences in height or weight between life-long vegetarians and people who became vegetarian at age 20 or older. Nor was there a significant difference in age at menarche between life-long vegetarian women and women who became vegetarian at age 20 or later.

Indeed, there is the possibility that fish consumption in childhood might actually be harmful, since another British study found that children who ate a lot of fish had twice the risk of suffering a stroke in adult life.

Well, how do we avoid heart disease if fish is both ineffective and unethical? The answer starts with nuts. Consuming about 30 grams of nuts per day at least five days a week was found by Gary Fraser to halve the risk of having a heart attack. No other single component of diet has been shown to be as effective. Forget fish, eat nuts daily along with legumes and whole foods generally. Avoid refined foods and eat plenty of fibre, particularly soluble fibre such as found in egg plant, okra, oats, barley, legumes and most vegetables and fruits.

If you are still not convinced that we make enough EPA and DHA of our own, and want consume more, then it is possible to buy capsules of them which have been manufactured from algae (which incidentally is where fish get them from too).

1. [Are dietary recommendations for the use of fish oils sustainable?](#)
2. [The Blue Zones](#)
3. [Legumes: the most important dietary predictor of survival in older people of different ethnicities.](#)
4. [Anatomy of health effects of Mediterranean diet: Greek EPIC prospective cohort study.](#)
5. [Long-chain omega-3 fatty acids, fish intake, and the risk of type 2 diabetes mellitus.](#)